

PKU–Zurich PhD Summer School

on Machine Learning for Macroeconomics and Finance

Reading List

Beijing, China · July 6–10, 2026

Before the Summer School (Pre-Reading & Pre-Coding)

Everyone should complete these two items before arriving in Beijing.

- **Reading — heterogeneous-agent models.** Dirk Krueger, *An Introduction to Macroeconomics with Household Heterogeneity* (lecture notes), 2026. [\[link\]](#)
Read Chapter 6, “The SIM in General Equilibrium,” carefully; use Chapter 4, “The SIM in Partial Equilibrium,” as a background reference.
- **Coding — neural networks for economic models.** Simon Scheidegger, tutorials on using neural networks to solve simple Brock–Mirman models (Jupyter notebooks).
Read and run both notebooks: Notebook 1 — [deterministic Brock–Mirman](#), and Notebook 2 — [stochastic Brock–Mirman](#).

Day 1 — Deep Learning and Heterogeneous-Agent Basics

Sergei Maliar (Santa Clara University)

- Maliar, Maliar & Winant (2021). *Deep learning for solving dynamic economic models*. Journal of Monetary Economics 122, 76–101. Code: github.com/marcmalier/deep-learning-euler-method-krusell-smith and github.com/QuantEcon/notebook-gallery. [\[link\]](#)
- Lepetyuk, Maliar & Maliar (2020). *When the U.S. catches a cold, Canada sneezes: A lower-bound tale told by deep learning*. Journal of Economic Dynamics and Control 117, 103926. [\[link\]](#)
- Maliar & Maliar (2021). *Deep learning classification: Modeling discrete labor choice*. Journal of Economic Dynamics and Control 135, 104295. [\[link\]](#)

Recommended background

- Goodfellow, Bengio & Courville (2016). *Deep Learning*. MIT Press (full text free online). [\[link\]](#)
- Judd (1998). *Numerical Methods in Economics*. MIT Press. [\[link\]](#)
- Maliar & Maliar (2014). *Numerical Methods for Large-Scale Dynamic Economic Models*. in K. Schmedders & K. Judd (Eds.), *Handbook of Computational Economics*, Vol. 3, Ch. 7, 325–477, Elsevier. [\[link\]](#)
- Aiyagari (1994). *Uninsured Idiosyncratic Risk and Aggregate Saving*. Quarterly Journal of Economics 109(3), 659–684. [\[link\]](#)
- Krusell & Smith (1998). *Income and Wealth Heterogeneity in the Macroeconomy*. Journal of Political Economy 106(5), 867–896. [\[link\]](#)

Day 2 — Deep Equilibrium Nets

Simon Scheidegger (HEC Lausanne) and Felix Kübler (University of Zurich)

- Scheidegger (2026). *Deep Learning for Solving and Estimating Dynamic Models in Economics and Finance*. arXiv:2605.14493 (lecture notes, with TensorFlow / PyTorch notebooks). Code: github.com/sischei/Deep_Learning_for_Solving_And_Estimating_Dynamic_Economic_Models. [\[link\]](#)
- Azinovic, Gaegauf & Scheidegger (2022). *Deep Equilibrium Nets*. International Economic Review 63(4), 1471–1525. Code: github.com/sischei/DeepEquilibriumNets. [\[link\]](#)
- Scheidegger & Bilonis (2019). *Machine learning for high-dimensional dynamic stochastic economies*. Journal of Computational Science 33, 68–82. [\[link\]](#)
- Chen, Didisheim & Scheidegger (2026). *Deep Surrogates for Finance: With an Application to Option Pricing*. Journal of Financial Economics 177, 104222. [\[link\]](#)
- Friedl, Kübler, Scheidegger & Usui (2021). *Deep Uncertainty Quantification: With an Application to Integrated Assessment Models*. Working paper, University of Lausanne. [\[link\]](#)
- Rasmussen & Williams (2006). *Gaussian Processes for Machine Learning*. MIT Press (full text free online). [\[link\]](#)

Day 3 — DeepHAM and Structural Reinforcement Learning

Yucheng Yang (University of Zurich), Benjamin Moll (LSE)

- Han, Yang & E (2026). *DeepHAM: A Global Solution Method for Heterogeneous Agent Models with Aggregate Shocks*. Quantitative Economics 17(2), 297–341. Code: github.com/frankhan91/DeepHAM. [\[link\]](#)
- Yang, Wang, Schaab & Moll (2025). *Structural Reinforcement Learning for Heterogeneous Agent Macroeconomics*. arXiv:2512.18892 (code coming soon). [\[link\]](#)
- Wibault, Forkel, Towers, Wibault, Duque, Whittle, Schaab, Yang, Wang, Osborne, Moll & Foerster (2026). *Recurrent Structural Policy Gradient for Partially Observable Mean Field Games*. International Conference on Machine Learning (ICML), 2026. Code: github.com/CWibault/mfax. [\[link\]](#)

Recommended background

- Moll (2026). *The Trouble with Rational Expectations in Heterogeneous Agent Models: A Challenge for Macroeconomics*. The Economic Journal 136(676), 1173–1216. [\[link\]](#)
- Zhao (2024). *Mathematical Foundations of Reinforcement Learning*. Springer Nature & Tsinghua University Press (esp. Ch. 6 on stochastic approximation / Robbins–Monro). [\[link\]](#)
- Sutton & Barto (2018). *Reinforcement Learning: An Introduction*. 2nd ed., MIT Press. [\[link\]](#)

Day 4 — Deep Learning and Continuous-Time Macro-Finance

Goutham Gopalakrishna (*University of Toronto, Rotman*)

- Gopalakrishna & Wu (2021). *ALIENs for Continuous Time Economies*. Swiss Finance Institute Research Paper No. 21-34. Docs: rotmanfinhub.github.io/deep-macrofin. [\[link\]](#)
- Raissi, Perdikaris & Karniadakis (2019). *Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations*. *Journal of Computational Physics* 378, 686–707. [\[link\]](#)

Recommended background

- Brunnermeier & Sannikov (2014). *A Macroeconomic Model with a Financial Sector*. *American Economic Review* 104(2), 379–421. [\[link\]](#)
- Achdou, Han, Lasry, Lions & Moll (2022). *Income and Wealth Distribution in Macroeconomics: A Continuous-Time Approach*. *Review of Economic Studies* 89(1), 45–86. [\[link\]](#)

Day 5 — Deep Learning for Macro-Finance and Beyond

Jonathan Payne (*Princeton University*)

- Gu, Laurière, Merkel & Payne (2024). *Global Solutions to Master Equations for Continuous Time Heterogeneous Agent Macroeconomic Models*. arXiv:2406.13726 (forthcoming, JPE: Macroeconomics). [\[link\]](#)
- Payne, Rebei & Yang (2024). *Deep Learning for Search and Matching Models*. Swiss Finance Institute Research Paper No. 25-05. [\[link\]](#)
- Gopalakrishna, Gu & Payne (2026). *Institutional Asset Pricing with Segmentation and Household Heterogeneity*. SSRN Working Paper. [\[link\]](#)